CA1
STAT2012

2026 Sem 2
TU850/2

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Introduction

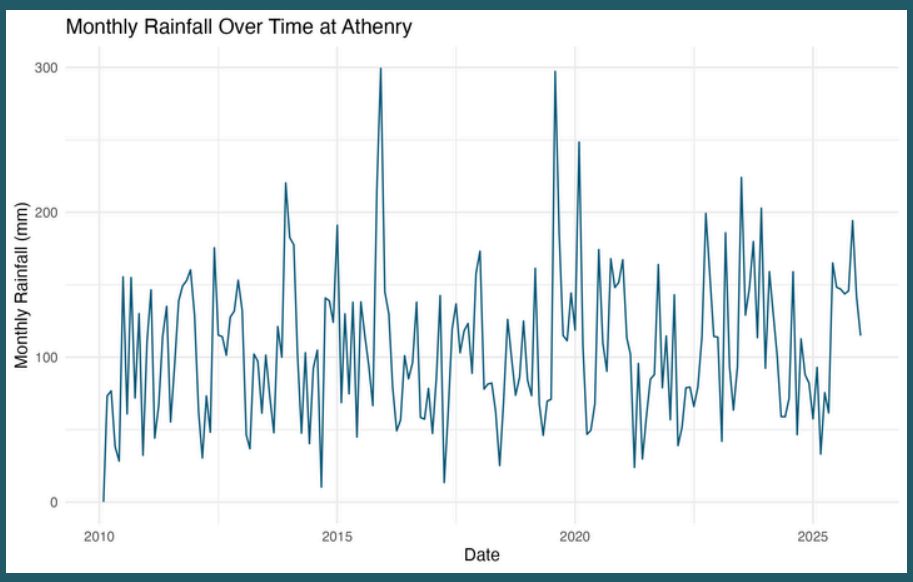
Rainfall and temperature are examples of weather patterns that change from place to place and time to time. When looking at historical weather data, it's simpler to see patterns and changes in the way the climate behaves. The dataset includes daily records of things like rainfall, the highest and lowest temperatures... The main goal is to look at the patterns of rainfall, figure out how likely it is to rain on a certain day, find the wettest month at the site and to compare daily temperatures to long-term averages. The language I chose to use was R with the tidyverse and forecast tools to complete the analysis. The dplyr package was used to change the data and the ggplot2 package was used to make graphs.

Dataset

This analysis was done on a dataset that had historical daily weather records from the Athenry weather station. The data provides information about the amount of rain, the highest temperature and the lowest temperature for each day.
The dataset was cleaned and changed before starting. The only variables that were kept were the date of the observation, the highest temperature, the lowest temperature and the amount of rain. The mean daily temperature was found by taking the average of the highest and lowest temperatures.
To help with the analysis, new variables were made, such as the year and month taken from the date. The daily rainfall readings were grouped by year and month to find out how much rain fell each month.
These changes made it possible to summarise and visualise the data to see how rainfall patterns and temperature changes over time.

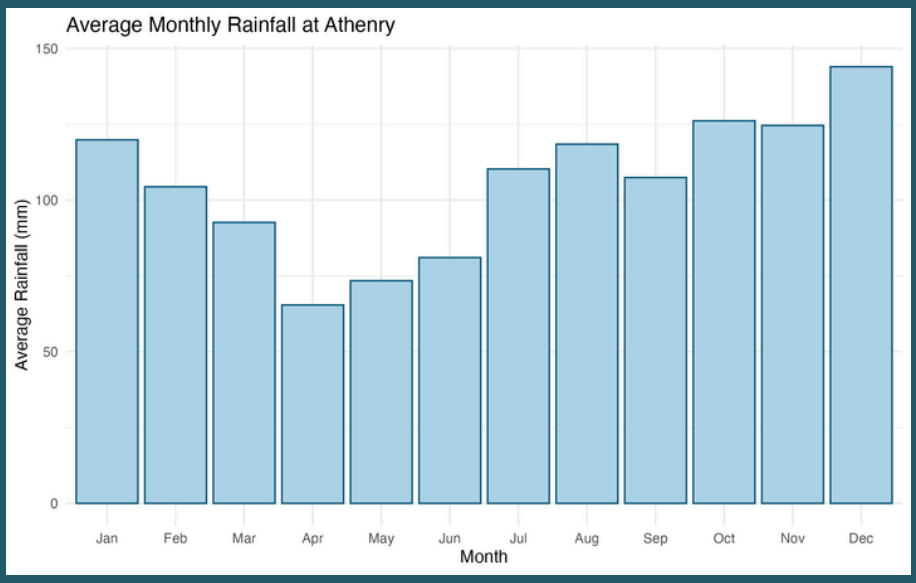
Rainfall Data Analysis

Monthly Rainfall Over Time



The diagram illustrates how much rain fell at the Athenry weather station each month over time. The line graph shows that the amount of rain that falls changes a lot from month to month. Some months have less rain than others, while others have a lot more. It also shows times when the amount of rain falls more than typical in a month.These peaks represent periods of wet conditions. Overall, the figure demonstrates that rainfall patterns fluctuate throughout the dataset and are not evenly distributed across time.

Distribution of Monthly Rainfall



The average monthly rainfall values were calculated to identify the wettest month at the Athenry weather station. The results show that December has the highest average monthly rainfall, with approximately 144 mm, followed by October (126 mm) and November (125 mm). The lowest average rainfall occurs in April, with approximately 65 mm.
This suggests that rainfall levels are generally higher during the autumn and winter months, while spring tends to be relatively drier. This seasonal pattern is typical for Ireland, where Atlantic weather systems often bring increased rainfall during the colder months.

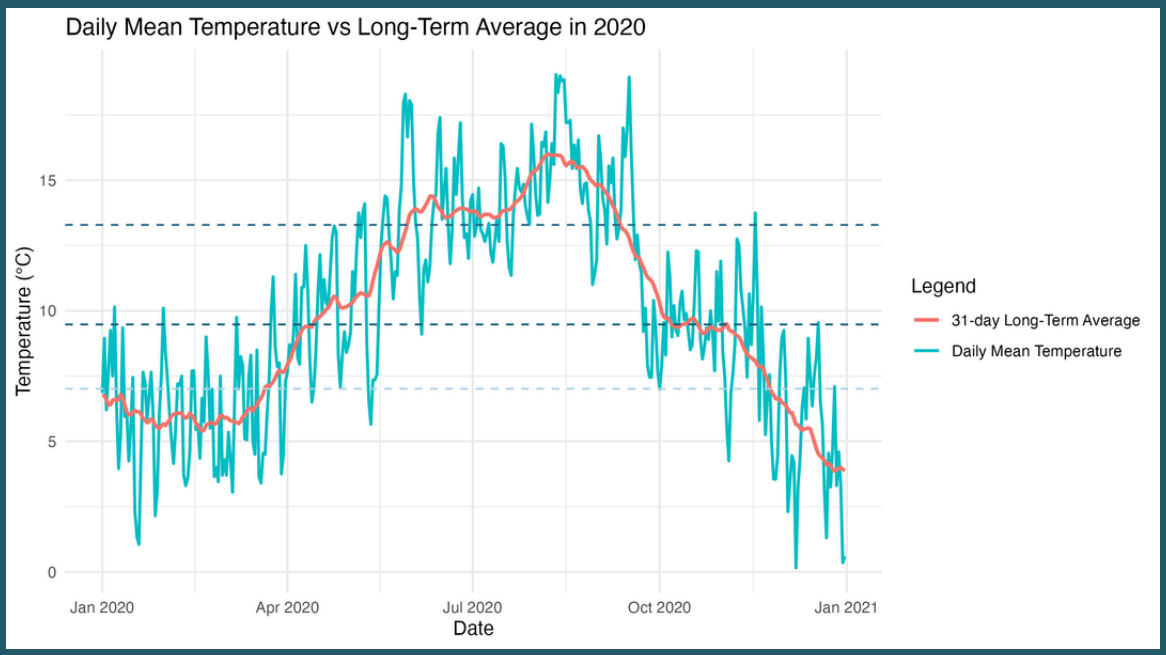
The probability of a rainy day was calculated using empirical probability, where a rainy day was defined as any day with rainfall greater than 0 mm. The probability was calculated as the proportion of rainy days relative to the total number of days in the dataset.

$$Empirical\ Probability = \frac{Number\ of\ Times\ Occured}{Total\ No.\ of\ times\ experiment\ performed}$$

The result shows that the probability of a rainy day is 0.73, which corresponds to approximately 73%. This means that a randomly selected day from the dataset has roughly a 73% chance of rainfall, highlighting the frequent occurrence of precipitation in Ireland's climate.

Temperature Data Analysis

The diagram shows the daily mean temperature for the year 2020 compared with a 31-day moving average representing the long-term temperature trend.
The moving average smooths short-term fluctuations in daily temperature and highlights the overall seasonal pattern. The figure shows that daily temperatures fluctuate around this long-term average. Periods where the daily temperature exceeds the moving average indicate warmer than usual conditions, while values below the average indicate cooler than usual conditions.



The first quartile (Q1) represents the temperature below which 25% of observations fall. The median (Q2) represents the middle value of the dataset, while the third quartile (Q3) represents the temperature below which 75% of observations fall.
In the temperature plot, horizontal lines representing these quartiles help identify typical and unusual temperature values. Temperatures above the third quartile indicate relatively warm days, while temperatures below the first quartile represent relatively cool days.

Conclusion

The investigation of the rain showed that the amount of rain changes a lot from month to month. Most months get considerable amounts of rain, while very wet months are less common. Using empirical probability, it was possible to figure out the average monthly rainfall and find out which month was the wettest - December.
The investigation of temperatures showed that daily temperatures vary around a long-term moving average, which shows how temperatures change with the seasons. Quartile analysis gave more information about how daily temperatures were spread out and helped find values that were normal and values that were uncommon.
The results show that both the temperature and the amount of rain at the Athenry weather station change clearly with the seasons. As a small reccomendation I think that future studies could analyse longer climate periods or compare multiple Irish weather stations to better understand regional climate variation.

Refrences

DataCamp, 2026. Statement of accomplishment: Data Manipulation with dplyr. Available at: https://www.datacamp.com/statement-of-accomplishment/course/98d0b7777be02ab368e3c86c4ab8570d8481bdbd?raw=1 (Accessed: 5 March 2026).

Wickham, H., 2024. Introduction to ggplot2. Available at: https://ggplot2.tidyverse.org/articles/ggplot2.html Accessed: 6 March 2026

RStudio (Posit), 2024. Data Visualisation with ggplot2 Cheat Sheet. Available at: https://rstudio.github.io/cheatsheets/html/data-visualization.html Accessed: 6 March 2026

R Graph Gallery, 2024. Data Visualisation with ggplot2. Available at: https://r-graph-gallery.com/ggplot2-package.html Accessed: 10 March 2026

DataCamp, 2026. Statement of accomplishment: Intermediate Data Manipulation with dplyr. Available at: https://www.datacamp.com/statement-of-accomplishment/course/c44110e6b0c766afc2bc8b2b91c56794e7db3a42?raw=1 Accessed: 11 March 2026

DataCamp, 2026. Statement of accomplishment: Introduction to Data Visualization with ggplot2. Available at: https://www.datacamp.com/statement-of-accomplishment/course/1a596496629eb4f5f0afa88b371de5f044e02f3d?raw=1 Accessed: 12 March 2026

GitHub link: https://github.com/bytelea/weather-data-statistics